Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	12 February 2026
Team ID	LTVIP2026TMIDS43267
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	4 Marks

Technical Architecture:

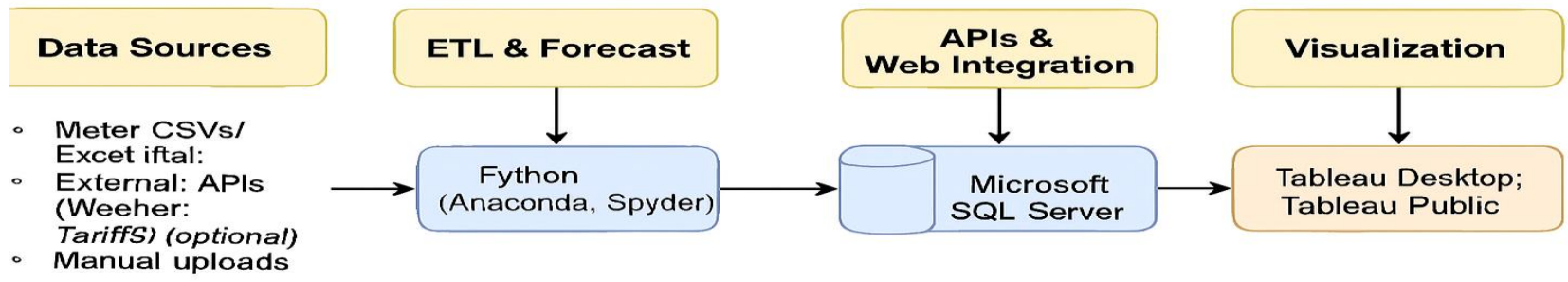

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Table -1: Components &Technology:

S.No	Component	Description	Technology
1	User Interface	Interactive dashboards and embedded visualizations	Tableau Desktop; Tableau Public
2	Application Logic-1 (ETL & Prep)	Data ingestion, cleaning, aggregation, extracts creation	Python (Anaconda) using pandas; Spyder / VS Code
3	Application Logic-2 (Forecasting / Analysis)	Time-series forecasting, seasonal analysis, anomaly detection	Python libraries: Prophet / statsmodels / scikit-learn
4	Application Logic-3 (Web Integration / APIs)	Serve processed datasets, embed dashboards, provide endpoints for visualizations	Flask (Python) developed in VS Code
5	Database	Primary storage for raw and cleaned time-series and sector metadata	Microsoft SQL Server
6	Cloud Database (optional)	Managed DB for production or sharing (if migrated)	Azure SQL Database; AWS RDS for SQL Server (optional)
7	File Storage	Local project files, CSVs, Tableau extracts, model artifacts	Local filesystem; Tableau .hyper extracts
8	External API-1 (optional)	Weather or external enrichment data for correlation analysis	OpenWeatherMap; Government open-data APIs
9	Machine Learning Model (optional)	Forecasting model artifacts saved for reuse	Prophet model files; job scripts in Python
10	Infrastructure (Dev / Prod)	Local dev and optional cloud hosting for Flask app and DB	Local machine for dev; Azure/AWS VM or App Service for deployment

Table-2: Application Characteristics:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	System should provide an easy-to-use and interactive Tableau dashboard interface
NFR-2	Security	User data and electricity records must be securely stored and encrypted
NFR-3	Reliability	System should provide accurate and consistent analysis results
NFR-4	Performance	Dashboard should load within acceptable response time (<3 seconds)
NFR-5	Availability	System should be available 24/7 with minimal downtime
NFR-6	Scalability	System should handle increasing data volume and multiple users efficiently